

6. (a) The work function of the metal rubidium is $3.62 \times 10^{-19} \text{ J}$. Explain what this statement means. [2]

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- (b) The minimum frequency of light that will eject electrons from a surface of work function ϕ is $\frac{\phi}{h}$. Explain in terms of photons why this is so. [3]

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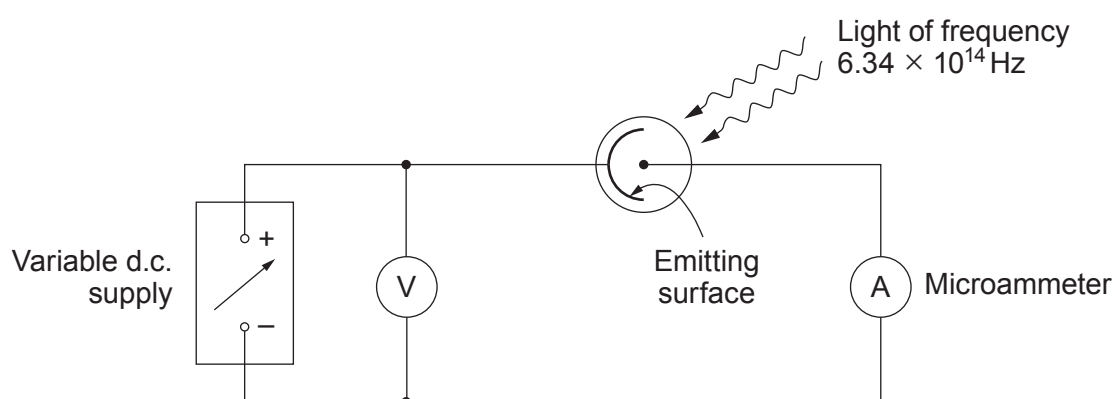
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- (c) A student has the task of determining whether or not the emitting electrode in a vacuum photocell might be made of rubidium. A light source of known frequency ($6.34 \times 10^{14} \text{ Hz}$) is provided. The student sets up the circuit shown.



The student shines the light on the emitting electrode and increases the pd from zero until the current has just fallen to zero. She records this pd (*the stopping pd*) and repeats the procedure, obtaining these results.

Stopping pd/V	0.381	0.366	0.388	0.373	0.371	0.380
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- (i) Calculate the mean value and the **absolute** uncertainty of the stopping pd, giving your answers to an appropriate number of significant figures. [3]

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- (ii) Evaluate whether or not the student should conclude that the emitting electrode might be made of rubidium. [Work function of rubidium = 3.62×10^{-19} J.] [4]

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